COURSE TITLE : PARTICLE TECHNOLOGY
COURSE CODE : 6073
COURSE CATEGORY : A
PERIODS/ WEEK : 5
PERIODS/ SEMESTER : 75
CREDIT : 5

TIME SCHEDULE

MODULE	TOPIC	PERIODS
1	Filtration and centrifugation	19
2	Size reduction and size separation	18
3	Sedimentation, floatation, agitation and mixing	19
4	Fluidization , Storage and Transportation of Solids, Liquids and Gases	19
TOTAL		**75**

COURSE OUTCOME :

SL.NO.	SUB	STUDENT WILL BE ABLE TO
1	1	Understand filtration as a solid liquid separation Recognize the selection criteria of filters
	2	Understand the process Centrifugation
2	3	Know the size reduction techniques Understand the working crushing equipments Understand the laws of crushing Understand the size separation and settling methods
3	4	Understand sedimentation methods and floatation methods Understand the working of agitation and mixing equipments
4	5	Understand Fluidisation
	6	Understand Storage and Transportation of Solids, Liquids and Gases

SPECIFIC OUTCOMES

MODULE– I

Filtration and Centrifugation

1.1.0 Understand filtration as a solid liquid separation

1.1.1 Define filtration
1.1.2 List the type of filters
1.1.3 Discuss the working of sand filter
1.1.4 Draw the sketch of plates and frames of filter press
1.1.5 Draw the sketch of plate and frames in the filtering operation
1.1.6 Discuss the working of plate and frame filter press
1.1.7 Explain the washing of plate and frame filter press

1.1.8 Explain the working of leaf filter
1.1.9 Explain the working of Moore filter
1.1.10 Explain the working of Rotary Drum Filter & Draw the sketch

1.2.0 Recognize the selection criteria of filters

1.2.1 List the factors on which rate of filtration depends
1.2.2 List the types of filter aids
1.2.3 State the function of filter aids in filtration
1.2.4 State the application of filter aids
1.2.5 List the types of filter cakes
1.2.6 Explain compressible and non-compressible filter cakes
1.2.7 Explain constant pressure filtration
1.2.8 Explain constant volume filtration
1.2.9 State the rate equation for constant pressure filtration
1.2.10 State term resistance of filter cake
1.2.11 Compute the time of filtration using the rate equation

1.3.0 Understand the process Centrifugation

1.3.1 List the classification of centrifuges
1.3.2 Draw the sketch of top driven taken centrifuge
1.3.3 Describe the working top driven batch centrifuge
1.3.4 Describe the working of bottom driven centrifuge
1.3.5 Draw the sketch of super centrifuge
1.3.6 Describe the working of super centrifuge
1.3.7 Draw the sketch of Disc centrifuge
1.3.8 Describe the working of disc centrifuge

MODULE – II

Size Reduction

2.1.0 Know the size reduction techniques

2.1.1 Generalize the important of size reduction
2.1.2 List the nature of materials to be crushed
2.1.3 Discuss closed circuit grinding
2.1.4 Discuss open circuit grinding
2.1.5 List the classification of crushing and grinding equipments

2.2.0 Understand the working of crushing equipments

2.2.1 Draw the sketch of Blake Jaw crusher
2.2.2 Explain the working of Blake Jaw crusher
2.2.3 Compare Blake and Dodge Jaw crusher
2.2.4 Draw the sketch of gyratory crusher
2.2.5 Explain the working of gyratory crusher
2.2.6 Explain crushing rolls
2.2.7 Draw the sketch of a smooth roll crusher
2.2.8 Define the angle of Nip of crushing rolls
2.2.9 Explain the theory behind the action of crushing rolls
2.2.10 Compute the angle of Nip
2.2.11 Draw the sketch of a single Roll crusher
2.2.12 Describe the working of a single roll crusher

- 2.2.13 Draw the sketch of a ball mill
- 2.2.14 Explain the working of ball mill
- 2.2.15 Compare ball and tube mill
- 2.2.16 List the factors influencing the size of the product of a ball mill
- 2.2.17 Derive an equation of critical speed of a ball mill
- 2.2.18 Compute critical speed

2.3.0 Understand the laws of crushing

- 2.3.1 Explain Rittingers law
- 2.3.2 State Kicks law
- 2.3.3 State Bond's law
- 2.3.4 Calculate power consumption using Rittingers law
- 2.3.5 Define particle shape
- 2.3.6 Define particle size
- 2.3.7 State mixed particle size
- 2.3.8 Define specific surface of mixture
- 2.3.9 Define average particle size
- 2.3.10 Define shape factor

Size Separation

2.4.0 Understand size separation

- 2.4.1 Compare the Tyler and U.S standard screens
- 2.4.2 List the types of screening equipment
- 2.4.3 Explain the working of grizzlies
- 2.4.4 Draw the sketch of a trommel
- 2.4.5 Explain the working of trommel
- 2.4.6 Differentiate between shaking and vibrating screen
- 2.4.7 Draw the sketch of vibrating screen

2.5.0 Understand settling methods

- 2.5.1 List the types of settling
- 2.5.2 Explain free settling
- 2.5.3 Derive Stokes law
- 2.5.4 Calculate terminal settling velocity using Stoke's law
- 2.5.5 Explain Hindered settling
- 2.5.6 Understand the mechanism of fluidization

MODULE III

Sedimentation, Agitation & Mixing

3.1.0 Understand sedimentation methods

- 3.1.1 Define sedimentation
- 3.1.2 Draw the sketches of various stages in Batch sedimentation
- 3.1.3 Explain Batch sedimentation
- 3.1.4 Explain continuous sedimentation
- 3.1.5 Discuss Kynch theory
- 3.1.6 Explain the determination of thickener area
- 3.1.7 Draw the sketch of Double cone classifier

3.1.8 State the principle of Jigging
3.1.9 Explain Tabling

3.2.0 Understand Floatation methods

3.2.1 Define flotation
3.2.2 Draw the flow sheet of floatation plant using rougher, scavenger and cleaner
3.2.3 Describe flotation equipment
3.2.4 Explain Magnetic separator

3.3.0 Understand the working of agitation and mixing equipments

3.3.1 Define Agitation
3.3.2 Distinguish between Agitation and Mixing
3.3.3 List the purpose of agitation
3.3.4 Draw the sketch of an agitation equipment
3.3.5 Explain the working of agitation equipment
3.3.6 Explain the types of Impellors
3.3.7 Draw the sketch of propellers
3.3.8 Explain paddles
3.3.9 Explain the flow pattern in agitated vessels
3.3.10 Explain the function of draft tube and Baffle
3.3.11 Explain the power consumption in agitated vessels
3.3.12 Define power Number.
3.3.13 Define Froude Number.
3.3.14 Calculation of power no, Froude no. and Reynold’s no. with respect to agitation and mixing
3.3.15 Explain dry mixer
3.3.16 Draw the sketch of Banbury mixer
3.3.17 Explain the working of Banbury mixer
3.3.18 Describe ‘V’ type mixer
3.3.19 Describe with a sketch kneading machine

MODULE – IV

Fluidisation, Storage and Transportation of Solids, Gases and Liquids

4.1.0 Fluidisation

4.1.1 Define fluidization
4.1.2 Discuss the mechanism of fluidization
4.1.3 Describe the characteristics of fluidization
4.1.4 Define minimum fluidization velocity
4.1.5 Define minimum bed height in fluidization
4.1.6 Define the term bed porosity
4.1.7 Discuss the effects of gas velocity on expansion of bed and pressure gradient
4.1.8 List the application of fluidization
4.1.9 Explain with a sketch fluidized bed coal conversion
4.1.10 Describe the fluidized bed oil conversion
4.1.11 Explain phthalic anhydride production
4.1.12 Explain the application of fluidized bed in catalytic cracking with a sketch
4.1.13 Describe application fluidized bed drying

4.2.0 Appreciate the storage of solids

4.2.1 Describe hoppers
4.2.2 Explain bins
4.2.3 Explain silos
4.2.4 Define angle of Repose
4.2.5 Discuss the points to be noted in the open storage of solids

4.3.0 Understand conveying methods and working of conveyors

4.3.1 Classification of conveyors
4.3.2 Draw the sketches of belt conveyor drives
4.3.3 Explain the belt conveyor construction
4.3.4 Draw the sketch of belt conveyor support belt conveyor idler
4.3.5 Explain belt conveyor take ups
4.3.6 Discuss belt conveyor feeders
4.3.7 Explain belt conveyor discharge methods
4.3.8 Explain chain conveyors
4.3.9 Discuss Apron conveyor
4.3.10 Describe the working of a screw conveyor
4.3.11 Draw the sketch of screw conveyor flight
4.3.12 Compare elevators and chain conveyors
4.3.13 Draw the sketch of a pneumatic
4.3.14 Explain the working of pneumatic conveyor

4.4.0 Know the storage of Liquid and gases

4.4.1 Define an atmospheric tank
4.4.2 Sketch some types of roofs of atmospheric storage tank
4.4.3 Define a gasholder
4.4.4 Define pressure vessel in storage of gases
4.4.5 Differentiate between bottles and pipelines in storage of gases

CONTENT DETAILS

MODULE I

Filtration and Centrifugation

Filtration as a solid, liquid separation and its application in industry. Classification of filters atmospheric, pressure and vacuum filters – field of application and constructional details, working and application of the following;

1. Sand filter – open – closed
2. Filter presses – plate and frame filter press, non-washing, open delivery, washing, closed delivery
3. Leaf filters – pressure and vacuum types – Moore filter.
4. Continuous filter – rotary drum – working cycle – methods of cake discharge – installation – horizontal pan filters – tilting pan filters – selection of filters – filter aids, their function and applications – pre-coating, filter media – types of filter mediums and its specific applications – properties of filter medium and selection.

Filter operation – effect of pressure – constant pressure and constant volume filtration. Rate equation for constant pressure filtration. Determination of constants in filtration equation – time of filtration – time of washing, simple problems. Specific resistance - compressible and non compressible cake.

Centrifugation:

Centrifugal force developed in centrifuges – classification of centrifuges – batch
– semi continuous, continuous – top driven – bottom driven – perforated solid bowl
– super centrifuges – operation – field of application.

MODULE II

Size Reduction
Nature of the materials to be crushed – hardness, structure, moisture content,crushing strength,
stickyness, soapyness – explosiveness
Types of crusting equipments, coarse crushers –Intermediatecrushers –fine grinders– open circuitgrinding – closed circuit grinding.
Laws of crushing – Kick's law – Rittingers law – Bonds law – Jaw crusher – gyratory – crusher – crushing rolls – angle of nip – simple problems – capacity, hammer mill – ball mill. – critical speed – simple problems – Raymond mill – tube mill.
Average particle size – specific surface of mixture, volume surface mean diameter – arithmetic mean diameter – mass mean diameter – shape factor.

Size Separation
Screens:TylerandU.S.standardscreensScreenanalysis:efficiency and capacity of screens
Types of screening equipment – grizzlies – trammels, shaking screens, vibrating screens
–separation of solids in liquids. Theory of settling, free and hindered setting – Stoke's law and its application,terminalvelocities–simpleproblem.

MODULE III

Sedimentation, Agitation and Mixing
Sedimentation separation in liquid medium – batch sedimentation – application of batch settling tests to design of continuous thickness – Kynch theory, determination of thickener area – equipments – double cone classifier – Dorr classifier –gravity continuous thickeners – elutriator jigging – tabling – light and dense medium separation based on difference in densities and its application.
Principle of froth flotation cells, froth floatation cells – simple flow sheet for floatation plant, magnetic separator

Agitation Mixing:
Purpose of agitation – agitation equipment – propellers, paddles and turbines.
Flow pattern in agitated vessels – prevention of swirling – draft tubes and baffles - their functions and effects.
Power consumption in agitated vessels – simple problems in determination of power consumption and design of mixers. Mixing of solids to solids. Ribbon blender – double cone, 'V' type mixes – pug mill.

MODULE IV

Fluidisation, Storage and Transportation of Solids, Storage of Gases and Liquidization:
Mechanism of fluidization – conditions for fluidization – batch fluidization – boiling effect – minimum porosity – bed height – minimum fluidization velocity – effect of gas velocity on expansion of bed and pressure gradient. Characteristics of gas solid fluidized systems – uses of fluidized bed systems – chemical reactors (catalytic and non catalytic) physical contacting drying – classification and conveying

Storage and Transportation of Solids, Gases and Liquids
Storage of solids – Hoppers – bins – angle of repose. Devices of for discharge of solids – Openstorage of solids
Conveyor types – Belt conveyor – Chain conveyor – Scraper conveyor – Apron conveyor – Bucket conveyors – Bucket elevators – Screw conveyors – Pneumatic conveyors – Pneumatic conveying system
Auxiliary equipments – Field of application of the above conveyors – Construction and working.
Storage of liquid – storage tanks, Storage of volatile liquids – floating roof, Storage of gases: Horton sphere – Wet and Dry specifications and codes for gas storage – Safety precautions.

REFERENCE

Waklter.L.Badger & Julius.T.Banchero	-	Introduction to chemical engineering
Warren.L.McCabe & Julian.C.Smith	-	Unit Operations
P.Chathopadhyay	-	Unit Operations of Chemical engineering. Vol 1
J.M.Coulson & Richardson	-	Chemical Engineering vol.II
Perry's	-	Chemical Engineers Hand Book